

HARSH CHANDRAMANIA

[Github](#): Spamziesagcan

[Linkedin](#): Harsh Chandramania

TIME SERIES ANALYSIS

- **Characteristics of time series data:** Time series data is a sequence of observations collected over time, typically at regular intervals. It's widely used in fields like finance, economics, weather forecasting, and machine learning. Here are the key characteristics of time series data:

1. Time Dependency

- Observations are ordered chronologically, and the order matters.
- Past values can influence future values (i.e., temporal correlation).
- Case scenario: Stock prices depend on their previous day's performance.

2. Trend

- Represents the long-term movement in the data.
- Could be increasing, decreasing, or constant over time.
- Case scenario: Population growth in a city over decades.

3. Seasonality

- Refers to repeating patterns or cycles at regular intervals (e.g., daily, monthly, yearly).
- Case scenario: Retail sales increase during holidays.

4. Cyclic Behavior

- Patterns that occur irregularly over time (not fixed like seasonality).

- Usually influenced by economic conditions, business cycles, etc.
- Case scenario: Economic booms and recessions affecting GDP.

5. Stationarity

- A time series is stationary if its statistical properties (mean, variance, autocorrelation) do not change over time.
- Many time series models (like ARIMA) require the data to be stationary.
- Case scenario: Monthly returns of a stable mutual fund.

6. Noise

- Refers to random variation in the data.
- Represents the unpredictable or unexplained part of the series.
- Case scenario: Daily sales data affected by random weather changes.

7. Autocorrelation

- Measures how related a current value is to its past values.
- Helps in identifying lags that affect the series.
- Case scenario: Temperature readings that depend on previous days.

8. Granularity

- The time interval between observations (e.g., hourly, daily, weekly).
- Granularity affects the type of patterns you can detect.
- Case scenario: Minute-wise electricity usage in a smart home.

9. Missing Values

- Real-world time series data often has gaps or missing entries.
 - Must be handled properly for accurate modeling.
 - Case scenario: Missing sensor readings due to power outages.
-

TIME SERIES DECOMPOSITION

- Trend: Long-term direction (upward/downward).
- Seasonality: Repeating patterns at fixed intervals.
- Cyclic: Repeating patterns but no fixed intervals.
- Residuals/Noise: Random fluctuations.

TYPES OF DECOMPOSITION

- **Additive:**
 - Adding components to retrieve time series data.
 - Case scenario: Website traffic that has a consistent upward trend, regular weekly peaks, and random fluctuations.
 - **Multiplicative:**
 - Multiplying components(when increasing) to retrieve time series data.
 - Case scenario: Monthly sales that increase in magnitude over time due to expanding customer base.
 - **STL decomposition using LOESS:**
 - Locally estimated scatterplot smoothing
 - Handling outliers in time series data
 - Showing actual seasonal patterns
 - Only handle additive models
 - Case scenario: Air passenger traffic data with both trend and seasonality, needing outlier handling.
-

STATIONARITY

- Constant over time - mean
 - variance
 - auto correlation

Need of Stationarity :

- Makes prediction easier because it assumes same statistical properties throughout the time.
- Forecasting models require stationary data.
- Case scenario: ARIMA models for predicting future oil prices require stationary data.

Types of Stationarity :

- **Weak Stationarity:**
 - Constant mean, variance and auto-correlation.
 - Joint distribution changes over time.
 - Useful for different forecasting models and short period of time.
 - Case scenario: Useful for short-term temperature prediction.
- **Strict Stationarity:**
 - Exhibits properties of weak stationarity.
 - Joint distribution remains unchanged when shifted along any time period.
 - Useful for modeling entire distribution of data.
 - Case scenario: Necessary for modeling the full distribution of financial returns.

Testing for Stationarity :

- **Weak Stationarity:**
 - **ADF Test:** Checks presence of unit root (non stationary trend)
 - **Null Hypothesis:** Data has a unit root.
 - **Alternate Hypothesis:** Data is stationary.

- **Decision criteria:**
 - $p < 0.05$ - stationary
 - ADF statistic < critical value - reject null hypothesis.
 - **Case scenario:** Applied to check if a sales trend is significant.
-
- **KPSS Test:**
 - Fits a constant mean model on the data
 - Measures the variance of the cumulative sum of residuals.
 - **Null Hypothesis:** Stationary
 - **Alternate Hypothesis:** Non-stationary
 - **Decision criteria:** KPSS statistic > critical value - reject null hypothesis.
 - **Case scenario:** Used to confirm stationarity after differencing.
-
- **Strict Stationarity:**
 - **KS Test:** Compares the cumulative distribution functions of two samples.
 - **Decision criteria:** $p > 0.05$ - no difference in distribution.
 - **Case scenario:** Used in simulations to compare distributions over time.

TECHNIQUES TO CONVERT NON-STATIONARY DATA TO STATIONARY

1. Differencing:

- Subtract current value from the previous value. Helps in removing linear trends.
- Case scenario: Removing a rising trend from monthly revenue.

2. Transformation: Stabilizes the variance of time series data. Types of transformation:

- **Log transformation:**
 - Takes log of all values to compress them.
 - Helps in multiplicative effects.
 - Case scenario: Stabilizing variance in exponential data like online followers.
 - **Power transformation:**
 - Takes power, like square root, to compress large values or expand small values.
 - Case scenario: Reducing skew in rainfall data.
 - **Box-Cox transformation:**
 - Combination of log and power transformation.
 - Case scenario: Handling non-linearity in web traffic data.
- 3. De-Trending:**
- Removing trend components by subtracting a trend line.
 - Case scenario: Isolating seasonal effect in retail sales.
- 4. Seasonal Adjustment:**
- Removing seasonal components,
 - Case scenario: Removing quarterly effects from GDP to analyze true growth.
-

DATA PREPARATION TECHNIQUES

- **Handling Missing Values:**
 - **Forward Fill:**
 - Forward fill replaces missing values with the most recent previous non-missing value.
 - Case scenario: Sensor data streams (e.g., temperature sensors) where the last reading is likely to hold until the next update.

- **Backward Fill:**
 - Backward fill replaces missing values with the next available non-missing value.
 - Case scenario: Filling missing survey data before a known event (e.g., weather conditions before a storm).
- **Linear Interpolation:**
 - Linear interpolation fills in missing values by drawing a straight line between the known values before and after the missing point. It assumes the change between those points is linear (steady and smooth).
 - Case scenario: Estimating missing hourly stock prices during system downtime (if price change is gradual).
- **Model-Based Imputation:**
 - Model-based imputation uses a predictive model (like linear regression, KNN, or time series models) to estimate missing values based on patterns in the data — either from:
 - Other time steps (temporal features)
 - Other related variables (multivariate context)
 - Case scenario: Predicting missing vital signs (e.g., heart rate) using previous readings and other patient features.
- **Outlier Detection & Treatment:**
 - **Z-Score / IQR Method:**
 - The Z-score measures how far a data point is from the mean, in terms of standard deviations.
 - Case scenario: Spotting unusually high transaction values in a customer's purchase history.
- **Resampling:**
 - **Aggregating Data:**
 - Aggregating data means reducing the frequency of your time series by combining multiple lower-frequency points into a higher-level one (e.g., from daily → weekly or hourly → daily).

- Case scenario: Converting minutely website traffic logs into daily summaries for trend analysis.
 - **Upsampling Interpolation:**
 - Upsampling increases the time frequency (e.g., from daily → hourly), often requiring interpolation to fill in the gaps between known values.
 - Case scenario: Filling missing timestamps in a high-frequency trading dataset.
 - **Feature Engineering:**
 - Creating lag features (e.g., yesterday's value)
 - Rolling windows (mean, std)
 - Time-based features (day of week, month, holidays)
 - Trend/seasonal indicators
 - Case scenario: In retail sales forecasting, you create features like "days since last promotion", "rolling 7-day average sales", and "is_holiday" to enrich the model's understanding.
-

TIME SERIES FORECASTING MODELS

1. Univariate:

- **Autoregression (AR):**
 - Uses past values of its time series to predict the future.
 - Order of AR model is denoted by (p), where p = no. of lagged observations.(observations considered to predict the future).
 - Case scenario: Predicting next day's temperature using past 5 days.
- **Moving Average (MA):**

- Uses the past error terms (residual).
 - Order of MA model is denoted by (q), where q = no. of lagged error terms.
 - Case scenario: Forecasting financial returns based on residuals.
- **Autoregressive Moving Average (ARMA):**
 - Combines AR & MA models for forecasting.
 - Captures both temporal dependencies & error terms.
 - Order of ARMA model is denoted by (p,q), where p = no. of lagged observations and q = no. of lagged error terms.
 - Case scenario: Combining both in electricity demand prediction.
 - **Autoregressive Integrated Moving Average (ARIMA):**
 - I (Integrated) part involved differencing order.
 - Order of ARIMA model is denoted by (p,d,q), where d = differencing order.
 - Case scenario: Forecasting airline passengers.
 - **Seasonal Autoregressive Integrated Moving Average (SARIMA):**
 - Useful with data having seasonality.
 - Order of SARIMA model is denoted by (p,d,q)(P,D,Q)m, where (P,D,Q) = order of seasonal part and m = time period.
 - Case scenario: Seasonal demand prediction for fashion items.

2. Multivariate:

Uses multiple time series & its dependencies to predict future values.

- **Vector Autoregression (VAR)**
 - A generalization of AR (Autoregression) to multiple time series. Each variable is modeled using its own past values and those of other variables.
 - Case scenario: Macroeconomic forecasting — model how GDP, inflation, and interest rates evolve together over time.
- **Vector Moving Average (VMA):**
 - Like MA in univariate time series, but for multiple variables each variable depends on past error terms of all variables.

- Case scenario: Shock propagation in markets — understand how unexpected events (shocks) in one variable (e.g., oil prices) affect others (e.g., airline stock prices).
- **Vector Autoregressive Moving Average (VARMA):**
 - A combination of VAR and VMA — each variable depends on past values and past errors of all variables.
 - Case scenario: Electricity grid forecasting — model how demand, generation, and price interact with each other over time, incorporating both historical patterns and random disturbances.
- **Vector Autoregressive Integrated Moving Average (VARIMA):**
 - Like VARMA, but also includes differencing (Integration) to make non-stationary data stationary — useful when trends exist in the multivariate series.
 - Case scenario: Inflation-adjusted forecasting — when forecasting exchange rates, stock indices, or consumer prices that show long-term trends, and need differencing before modeling their interaction.

3. Advanced Forecasting Models:

- **Seasonal AutoRegressive Integrated Moving Average with eXogenous variables (SARIMAX):**
 - An extension of SARIMA that includes exogenous (external) variables to improve forecasts.
 - $SARIMA(p, d, q)(P, D, Q, s)$: handles seasonality
 - Case scenario: Forecasting product sales influenced by external factors like promotions, holidays, or ad spend.
- **Prophet (Developed by Facebook):**
 - An open-source tool designed for business users and analysts, automatically models trend, seasonality, and holidays.
 - Handles missing data and outliers
 - Detects change points in trends
 - Great for daily/monthly data with strong seasonal cycles
 - Case scenario: Forecasting website traffic, sales during holidays, or social media engagement.

- **Generalized Autoregressive Conditional Heteroskedasticity (GARCH):**
 - A model used to estimate volatility over time, especially where variance is not constant (heteroskedasticity).
 - Captures volatility clustering (periods of high/low volatility)
 - Widely used in finance
 - Case scenario: Modeling stock market volatility, crypto price variance, or value-at-risk (VaR).
-

SMOOTHING METHODS

- **Importance of Smoothing:**
 - Noise reduction
 - Good for forecasting high fluctuated time series data
 - Trend identification
- **Moving Average:**
 - **Simple Moving Average (SMA):**
 - Case scenario: Reducing noise in weekly stock prices.
 - **Weighted Moving Average (WMA):**
 - Weights are assigned to the values which helps recent observations to have more influence on the prediction values.
 - Case scenario: Emphasizing recent trends in temperature forecasts.
 - **Exponential Moving Average (EMA):**
 - Smoothing factor = $2/n+1$, where n = window size.
 - Case scenario: Tracking short-term trends in cryptocurrency prices.

- **Exponential Smoothing:**

- If smoothing factor - high value - current observations are assigned more weights
- If smoothing factor - low value - previous observations are assigned more weights
- $0 < \text{smoothing factor} < 1$

- **Single Exponential Smoothing (SES):**

- Suitable for data with no trend or seasonality.
- Case scenario: Predicting demand for a stable product.

- **Double Exponential Smoothing (DES)(Holt's Linear):**

- Suitable for data with trend.
- Case scenario: Modeling upward-trending website visits.

- **Triple Exponential Smoothing (TES)(Holt's Winters):**

- Suitable for data with seasonality.
- Case scenario: Forecasting seasonal product sales like winter jackets.

MODEL SELECTION

- **Akaike Information Criterion (AIC):**

- AIC measures how well a model fits the data while penalizing complexity. It helps in selecting the best model from a set of candidates.
- Lower AIC = better model
- Case scenario: Choosing between ARIMA models (e.g., ARIMA(1,1,1) vs ARIMA(2,1,0)) to forecast product sales.

- **Bayesian Information Criterion (BIC):**

- Similar to AIC, but penalizes complexity more heavily. This makes it more conservative and often favors simpler models.
 - Lower BIC = better model, but usually simpler than the AIC-preferred one.
 - Case scenario: When forecasting long-term trends in energy usage, where simpler models are more stable and interpretable.
-

MODEL DIAGNOSTICS

- **Residual Analysis:**

- Residuals (actual – predicted) should behave like white noise: no trend, zero mean, constant variance, no autocorrelation
- Plot residuals and histogram/QQ plot to visually assess
- Case scenario: Forecasting daily sales of a fashion retailer using an ARIMA model.

- **Auto-Correlation Function (ACF) / Partial Auto-Correlation Function (PACF):**

- If residuals show autocorrelation → model missed some patterns
- Should ideally be near zero beyond lag 0
- Case scenario: After modeling weather temperature data, ACF plots show residual autocorrelation → suggesting you need more AR/MA terms.

- **Ljung-Box Test:**

- Statistical test for autocorrelation in residuals
- $p > 0.05 \rightarrow$ residuals are independent

- Case scenario: Validating your model on server load forecasting; a low p-value in Ljung-Box means the model's residuals are autocorrelated → not reliable for alerts.

- **Normality Tests:**

- Test if residuals are normally distributed (important for confidence intervals)
- Case scenario: You want to construct confidence intervals for financial forecasts. If residuals are not normal, intervals might be inaccurate, prompting transformations or model changes.

MODEL EVALUATION METRICS

- **Mean Absolute Error (MAE):**

- Measures the magnitude of errors in prediction without considering directions.
- $MAE = \frac{1}{n} \sum |Y(t) - Y'(t)|$, where $t = 1$ to n , $Y(t)$ = actual value & $Y'(t)$ = predicted value.
- Treats all errors (positive or negative) equally.
- Less sensitive to outliers.
- Case scenario: Evaluating hospital patient inflow forecast.

- **Mean Squared Error (MSE):**

- Measures the average of square of errors.
- Emphasizes more on large errors.
- $MSE = \frac{1}{n} \sum (Y(t) - Y'(t))^2$, where $t = 1$ to n , $Y(t)$ = actual value & $Y'(t)$ = predicted value.
- Sensitive to outliers.
- Useful for detecting large errors.
- Case scenario: Measuring error in stock price prediction.

- **Root Mean Squared Error (RMSE):**

- $RMSE = \sqrt{MSE}$
- Useful in fields where smallest of errors can cause a lot of damage.
- Case scenario: Accuracy of weather forecasts.

- **Mean Absolute Percentage Error (MAPE):**

- $MAPE = \frac{1}{n} \sum |(Y(t) - Y'(t)) / Y(t)| \times 100$, where $t = 1$ to n , $Y(t)$ = actual value & $Y'(t)$ = predicted value.
- Case scenario: Forecast accuracy for percentage-based business KPIs.